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D3 – V6 ASSIGNMENT

# WAREHOUSE ROBOTICS: CONVEYOR BELT DYNAMICS

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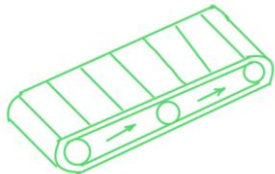
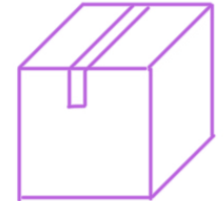
# CONTEXT

The project analyses the integration of manipulator robots with conveyor belt systems



Robot: mobile or fixed manipulator robots that move around the warehouse, performing pick/place/load conveyor operations

Package: items that are manipulated by robots and transported by conveyor belts



Conveyor belt: system useful within the warehouse to transport packages from one location to another, saving the movement of robots



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# PDDL

# Q1: PDDL

## DISCRETE APPROXIMATION

continuous conveyor dynamics are discretized into adjacent steps using the conveyor-step action

## SEPARATION OF ROLES

clear distinction between controlled actions (move, pick, place) and conveyor belt advancement actions (conveyor-step at 0 cost)

## RESOURCE OPTIMIZATION

using the total-cost function to evaluate whether the use of the belt is advantageous compared to manual transport with robots



# Q1: PREDICATES & ACTIONS

MOVE

PICK

MOVE

LOAD-

CONVEYOR

CONVEYOR-  
STEP

```
(:types
  location
  robot
  package
)
```

```
(:predicates
  (connected ?from - location ?to - location) 1
  (belt-connected ?from - location ?to - location)
  (robot-at ?r - robot ?l - location) 5 1 1
  (package-at ?p - package ?l - location) 3 3 3
  (holding ?r - robot ?p - package) 2 2 2
  (handempty ?r - robot) 2 2 2
  (on-belt ?p - package) 3 1 1
  (load-zone ?l - location) 1
  (unload-zone ?l - location) 2
  (delivery-zone ?l - location) 1
  (belt-free ?l - location) 2 2 2
)
```

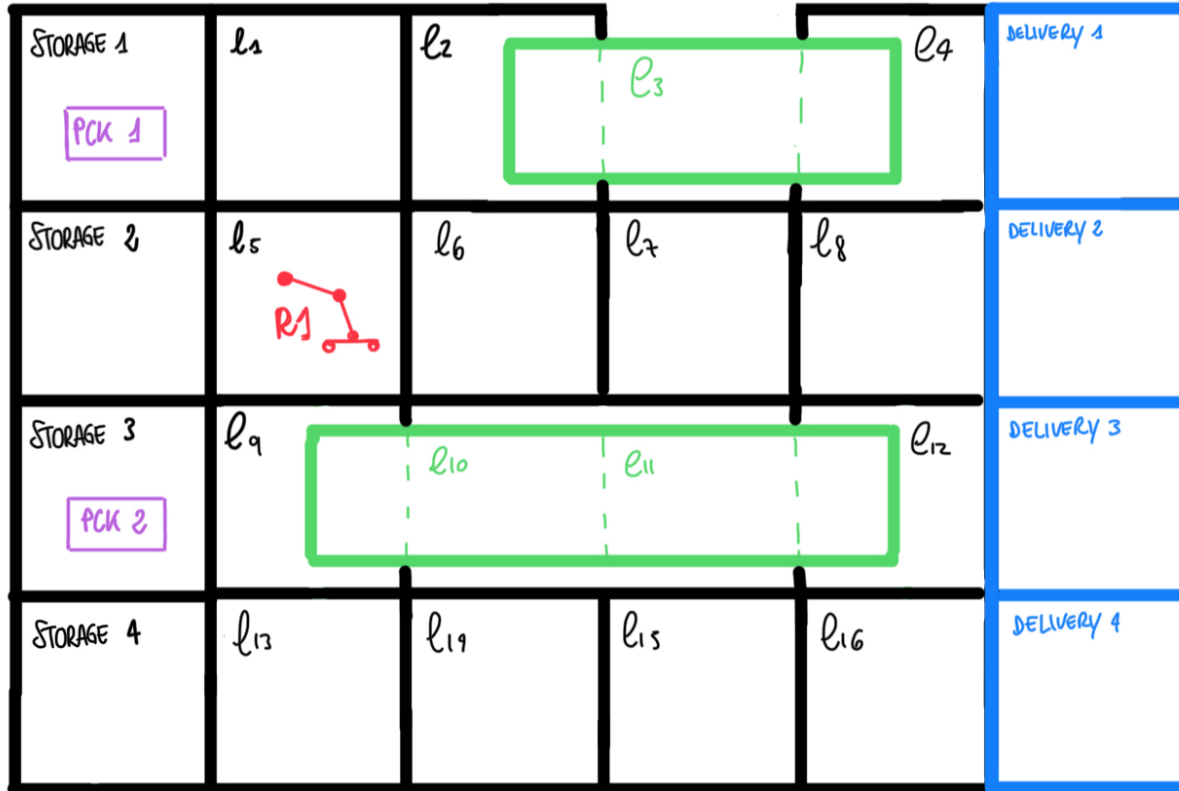
RETRIVE

MOVE

PLAC  
E



# PROBLEM 1: OPTIONAL USE OF THE CONVEYOR BELTS



•robot  $r_1$  can reach all zones

•the conveyor-belts are an efficient alternative, it does not increase the metric

## PLAN FOUND BY THE PLANNER R ENHSP

```

| move r1 l5 l9
| move r1 l9 storage3
| pick pck2 r1 storage3
| move r1 storage3 l9
| move r1 l9 l5
| move r1 l5 l6
| move r1 l6 l7
| move r1 l7 l8
| move r1 l8 delivery2
| move r1 delivery2 delivery3
| move r1 delivery3 delivery4
| place pck2 r1 delivery4
| move r1 delivery4 delivery3
| move r1 delivery3 delivery2
| move r1 delivery2 l8
| move r1 l8 l7
| move r1 l7 l6
| move r1 l6 l5
| move r1 l5 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l5
| move r1 l5 l6
| move r1 l6 l7
| move r1 l7 l8
| move r1 l8 l4
| move r1 l4 delivery1
| place pck1 r1 delivery1

```

```

| move r1 l5 l9
| move r1 l9 l5
| move r1 l5 storage2
| move r1 storage2 storage1
| pick pck1 r1 storage1
| move r1 storage1 storage2
| move r1 storage2 l5
| move r1 l5 l9
| load-conveyor r1 pck1 l9
| conveyor-step pck1 l9 l10
| conveyor-step pck1 l10 l11
| move r1 l9 storage3
| pick pck2 r1 storage3
| move r1 storage3 l9
| move r1 l9 l13
| move r1 l13 l14
| move r1 l14 l15
| move r1 l15 l16
| move r1 l16 delivery4
| place pck2 r1 delivery4
| move r1 delivery4 delivery3
| move r1 delivery3 l12
| conveyor-step pck1 l11 l12
| retrieve r1 pck1 l12
| move r1 l12 l8
| move r1 l8 l4
| move r1 l4 delivery1
| place pck1 r1 delivery1

```



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## OPTIMAL PLAN

# COMPARATIVE LOG

PLAN  
FOUND BY  
THE  
PLANNER  
ENHSP

```
Plan-Length:29
Metric (Search):54.0
Planning Time (msec): 40
Heuristic Time (msec): 18
Search Time (msec): 37
Expanded Nodes:41
States Evaluated:68
Number of Dead-Ends detected:3
Number of Duplicates detected:52
```

```
Plan-Length:28
Metric (Search):44.0
Planning Time (msec): 209
Heuristic Time (msec): 156
Search Time (msec): 208
Expanded Nodes:3934
States Evaluated:4468
Number of Dead-Ends detected:1301
Number of Duplicates detected:7219
```

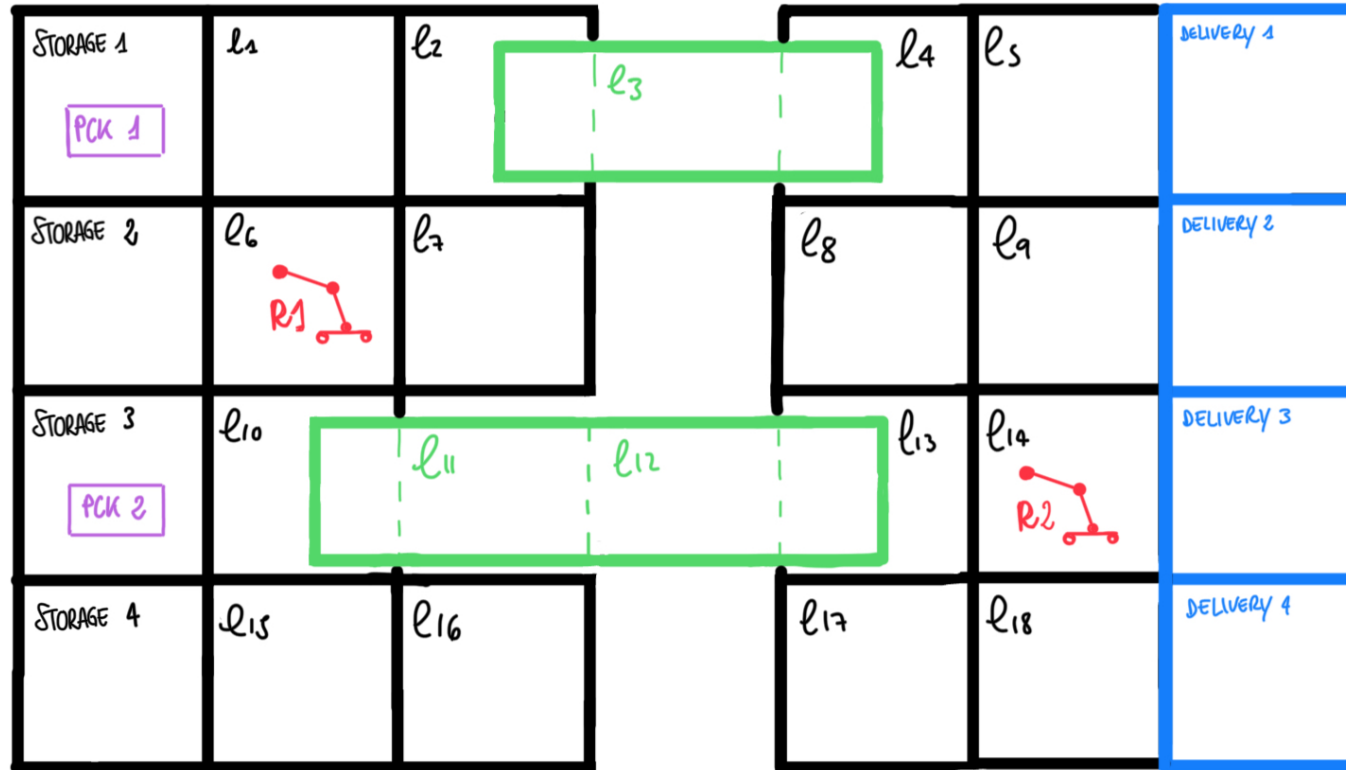
OPTIMAL  
PLAN

- the total plan length decreased from 29 to 28 actions
- the optimal plan achieved a strict cost reduction from 54.0 to 44.0 (-10.0 cost units), by replacing expensive robot moves with zero-cost conveyor steps
- the ENHSP planner found a satisfiable plan with 5.6x less time with respect to mathematically guarantee the absolute minimum cost
- the optimal solution explored 3,934 nodes and 4,468 states instead of 41 and 68 of the ENHSP planner respectively
- dead-ends rose from 3 to 1,301 and duplicates from 52 to 7,219 to delete all inefficient paths.





## PROBLEM 2: NECESSARY USE OF THE CONVEYOR BELTS



•two mobile manipulator robots operating in different areas

•conveyor-belt necessary to satisfy the plan

## PLAN FOUND BY THE PLANNER ENHSP

```

| move r1 l6 l10
| move r2 l14 l9
| move r2 l9 l5
| move r1 l10 storage3
| move r2 l5 l4
| pick pck2 r1 storage3
| move r1 storage3 l10
| load-conveyor r1 pck2 l10
| move r1 l10 l6
| move r1 l6 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l2
| load-conveyor r1 pck1 l2
| move r1 l2 l1
| conveyor-step pck2 l10 l11
| conveyor-step pck2 l11 l12
| conveyor-step pck2 l12 l13
| move r2 l4 l8
| move r1 l1 storage1
| move r2 l8 l13
| retrieve r2 pck2 l13
| move r2 l13 l14
| move r2 l14 delivery3
| move r1 storage1 l1
| move r1 l1 l2
| conveyor-step pck1 l2 l3
| conveyor-step pck1 l3 l4
| move r2 delivery3 delivery4
| place pck2 r2 delivery4
| move r2 delivery4 delivery3
| move r2 delivery3 delivery2
| move r2 delivery2 delivery1
| move r2 delivery1 l5
| move r2 l5 l4
| retrieve r2 pck1 l4
| move r2 l4 l5
| move r2 l5 delivery1
| place pck1 r2 delivery1

```

```

| move r1 l6 l10
| move r2 l14 l13
| move r1 l10 storage3
| move r2 l13 l8
| move r2 l8 l4
| pick pck2 r1 storage3
| move r1 storage3 l10
| load-conveyor r1 pck2 l10
| move r1 l10 l6
| move r1 l6 l1
| move r1 l1 storage1
| pick pck1 r1 storage1
| move r1 storage1 l1
| move r1 l1 l2
| load-conveyor r1 pck1 l2
| conveyor-step pck1 l2 l3
| conveyor-step pck1 l3 l4
| conveyor-step pck2 l10 l11
| retrieve r2 pck1 l4
| move r2 l4 l5
| move r2 l5 delivery1
| place pck1 r2 delivery1
| move r2 delivery1 delivery2
| move r2 delivery2 delivery3
| move r2 delivery3 l14
| move r2 l14 l13
| conveyor-step pck2 l11 l12
| conveyor-step pck2 l12 l13
| retrieve r2 pck2 l13
| move r2 l13 l17
| move r2 l17 l18
| move r2 l18 delivery4
| place pck2 r2 delivery4

```

## OPTIMAL PLAN

# COMPARATIVE LOG

PLAN  
FOUND BY  
THE  
PLANNER  
**ENHSP**

```
Plan-Length:38  
Metric (Search):58.0  
Planning Time (msec): 193  
Heuristic Time (msec): 134  
Search Time (msec): 192  
Expanded Nodes:2100  
States Evaluated:3644  
Number of Dead-Ends detected:95  
Number of Duplicates detected:10595
```

```
Plan-Length:33  
Metric (Search):48.0  
Planning Time (msec): 1045  
Heuristic Time (msec): 852  
Search Time (msec): 1041  
Expanded Nodes:30887  
States Evaluated:32867  
Number of Dead-Ends detected:25827  
Number of Duplicates detected:142569
```

OPTIMAL  
PLAN

- the total plan length decreased from 38 to 33 actions
- the optimal plan achieved a strict cost reduction from 58.0 to 48.0 (-10.0 cost units), by replacing expensive robot moves with zero-cost conveyor steps
- the ENHSP planner found a satisfiable plan with 5x less time with respect to mathematically guarantee the absolute minimum cost
- the optimal solution explored 30,887 nodes and 32,867 states instead of 2100 and 3644 of the ENHSP planner respectively
- dead-ends rose from 95 to 25827 and duplicates from 10595 to 142569 to delete all inefficient paths.

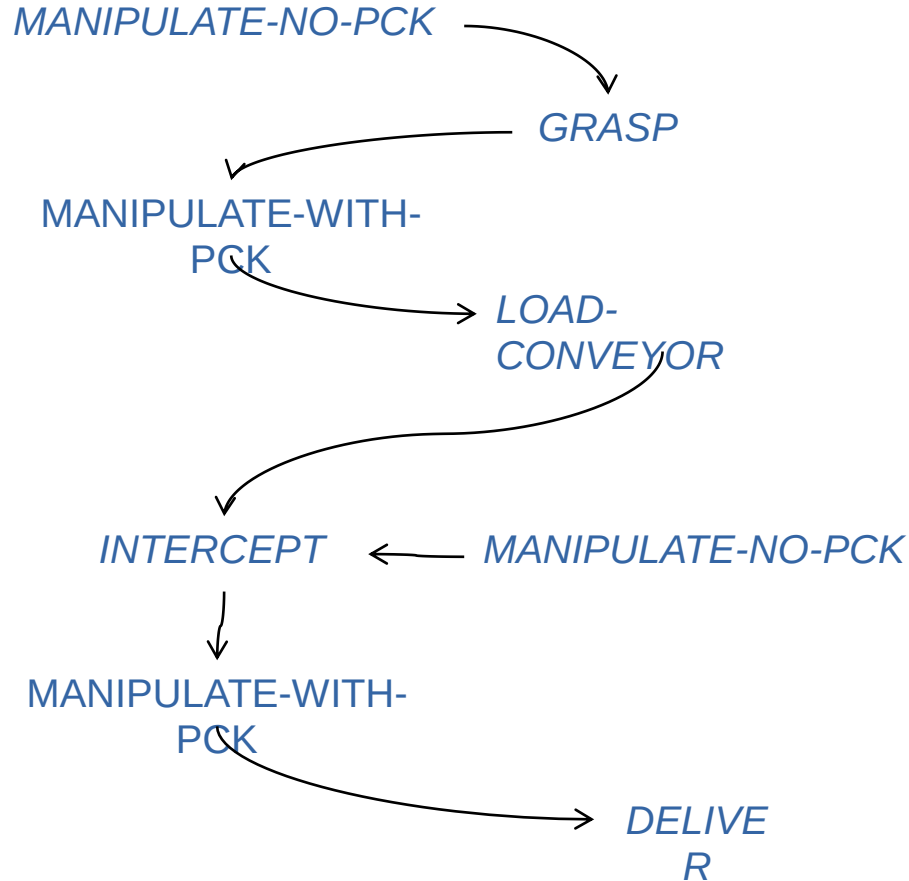


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## Q2: PREDICATES, FUNCTIONS, AND ACTIONS



```
(:functions
  (belt-progress ?p - package) 3
  (conveyor-speed ?c - conveyor)
  (busy-timer ?r - robot) 5 1\
  (total-cost) 6\
)
```

```
;; belt
(belt-segment ?c - conveyor ?from - location ?to - location) 1
(conveyor-entry ?c - conveyor ?l - location) 3
(conveyor-exit ?c - conveyor ?l - location) 4
(belt-free ?l - location) 3 4 3
(belt-running ?c - conveyor) 2
(conveyor-connected ?c1 - conveyor ?c2 - conveyor ?l1 - location ?l2 - location)

;; robot state
(robot-at ?r - robot ?l - location) 6 2 2
(robot-act-on ?r - robot ?l - location) 8
(holding ?r - robot ?p - package) 4 3 2
(handempty ?r - robot) 4 2 2
(busy ?r - robot) 7 4 3

;; package state
(package-at ?p - package ?l - location) 5 4 4
(on-belt ?p - package ?c - conveyor) 5 2 3

;; zones
(load-zone ?l - location) 3
(unload-zone ?l - location) 3
(delivery-zone ?l - location) 2
```

# Q2: PROCESSES

## TIME INTEGRATION

the native temporal variable `#t` is used  
to increase the position progress  
fluent based on conveyor velocity

## AUTONOMOUS MOVEMENT

package transit on conveyor belts is  
modeled using the continuous process  
**package-movement**

```
(:process package-movement
  :parameters (?p - package ?c - conveyor)
  :precondition (and (on-belt ?p ?c)
                    (belt-running ?c)
                    (< (belt-progress ?p) 1.0))
  :effect (increase (belt-progress ?p) (* #t (conveyor-speed ?c)))
)
```

## ROBOT TIMERS

independent continuous processes  
(**robot-timers**) handle internal clocks  
to track robot occupancy over time.

```
(:process robot-timers
  :parameters (?r - robot)
  :precondition (> (busy-timer ?r) 0.0)
  :effect (decrease (busy-timer ?r) (* #t 1.0))
)
```



# Q2: EVENTS

## TRANSITION

the event **advance-segment** triggers mathematically the exact moment a package progress reaches  $\geq 1.0$ , transferring it to the next conveyor segment (a location)

```
(:event advance-segment
:parameters (?p - package ?c - conveyor ?l1 - location ?l2 - location)
:precondition (and (on-belt ?p ?c)
                  (package-at ?p ?l1)
                  (belt-segment ?c ?l1 ?l2)
                  (belt-free ?l2)
                  (not (conveyor-exit ?c ?l1))
                  (>= (belt-progress ?p) 1.0)
                  )
:effect (and (not (package-at ?p ?l1))
             (package-at ?p ?l2)
             (belt-free ?l1)
             (not (belt-free ?l2))
             (assign (belt-progress ?p) 0.0)
             )
)
```



# Q2: EVENTS

## ROBOT EVENT

the event **free-robot** triggers mathematically the exact instant a robot finishes an operational task, automatically clearing its busy status so it can accept new planning commands.

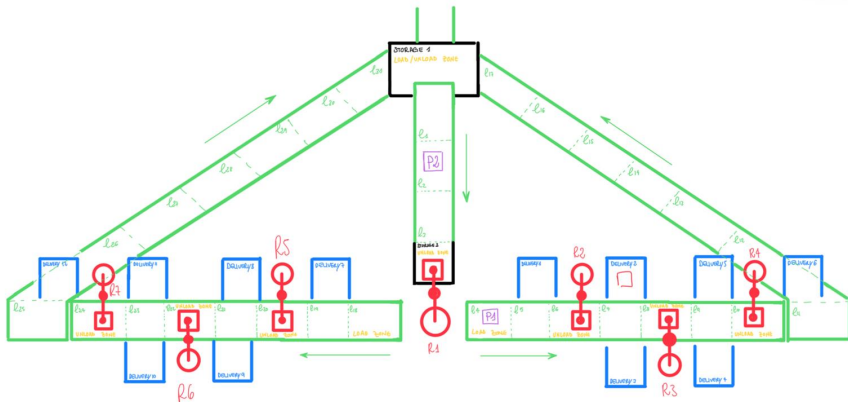
```
(:event free-robot
  :parameters (?r - robot)
  :precondition (and (busy ?r)
                    (<= (busy-timer ?r) 0.0)
                    (handempty ?r))
  :effect (and (not (busy ?r)))
)
```





# Q2: EVENTS

**TOPOLOGY INTERCONNECTION**  
network routing between different  
connected belts is autonomously  
dictated by the **reach-exit** event.

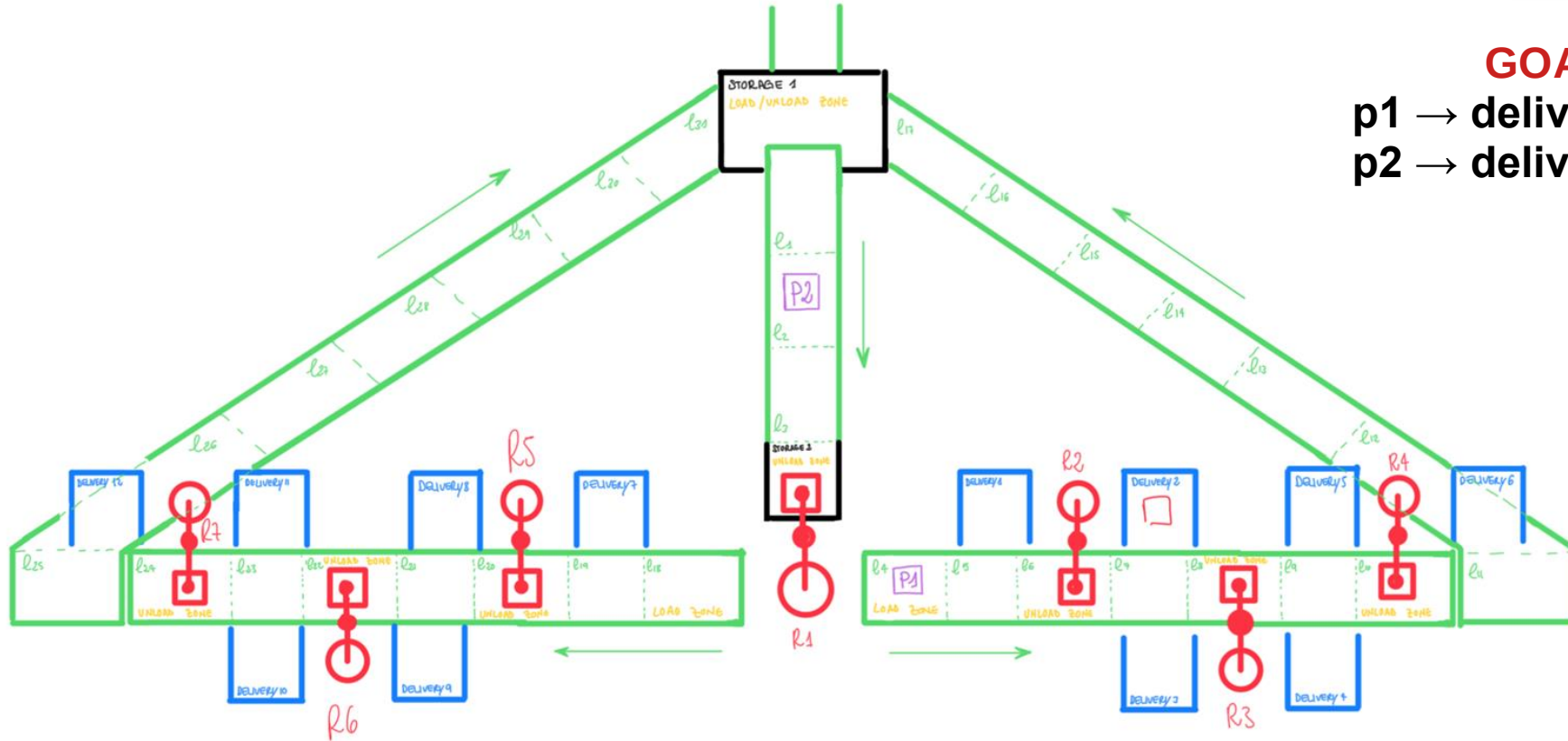


```
(:event reach-exit
  :parameters (?p - package ?cFrom ?cTo - conveyor ?from ?to - location)
  :precondition (and (on-belt ?p ?cFrom)
                     (package-at ?p ?from)
                     (conveyor-exit ?cFrom ?from)
                     (conveyor-entry ?cTo ?to)
                     (conveyor-connected ?cFrom ?cTo ?from ?to)
                     (belt-free ?to)
                     (>= (belt-progress ?p) 1.0))
  :effect (and (not (package-at ?p ?from))
               (package-at ?p ?to)
               (not (on-belt ?p ?cFrom))
               (on-belt ?p ?cTo)
               (belt-free ?from)
               (not (belt-free ?to))
               (assign (belt-progress ?p) 0.0))
)
```

# PROBLEM 1



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Found Plan:

```
0: -----waiting---- [4.0]
4.0: (intercept r2 p1 l6 cRight)
4.0: -----waiting---- [5.0]
5.0: (manipulate-with-package r2 p1 l6 delivery2)
5.0: -----waiting---- [6.0]
6.0: (grasp r1 p2 storage2 cMain)
6.0: -----waiting---- [7.0]
7.0: (manipulate-with-package r1 p2 storage2 l18)
7.0: (deliver r2 p1 delivery2)
7.0: -----waiting---- [9.0]
9.0: (load-conveyor r1 p2 l18 cLeft)
9.0: -----waiting---- [21.0]
21.0: (grasp r7 p2 l24 cLeft)
21.0: -----waiting---- [22.0]
22.0: (manipulate-with-package r7 p2 l24 delivery12)
22.0: -----waiting---- [24.0]
24.0: (deliver r7 p2 delivery12)
```

Plan-Length:43

Elapsed Time: 24.0

Metric (Search):75.0

Planning Time (msec): 52

Heuristic Time (msec): 34

Search Time (msec): 46

Expanded Nodes:34

States Evaluated:52

Number of Dead-Ends detected:4

Number of Duplicates detected:1

Computation Times:

- total planning time: 52s

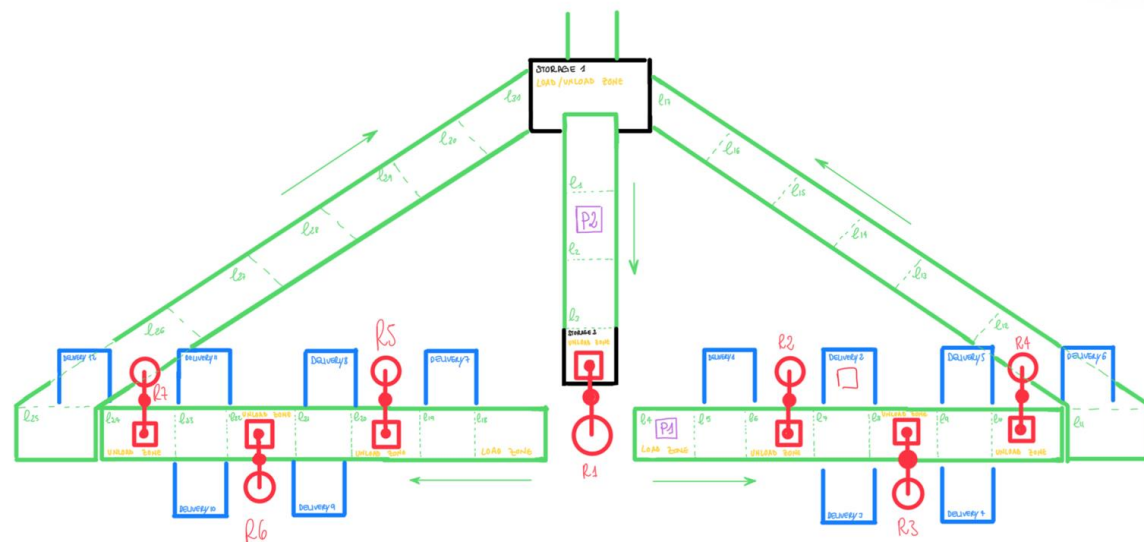
- time spent computing the heuristic node estimate: 34s

- time spent exploring the state space: 46s

- Only 4 Dead-Ends and 1 Duplicate detected



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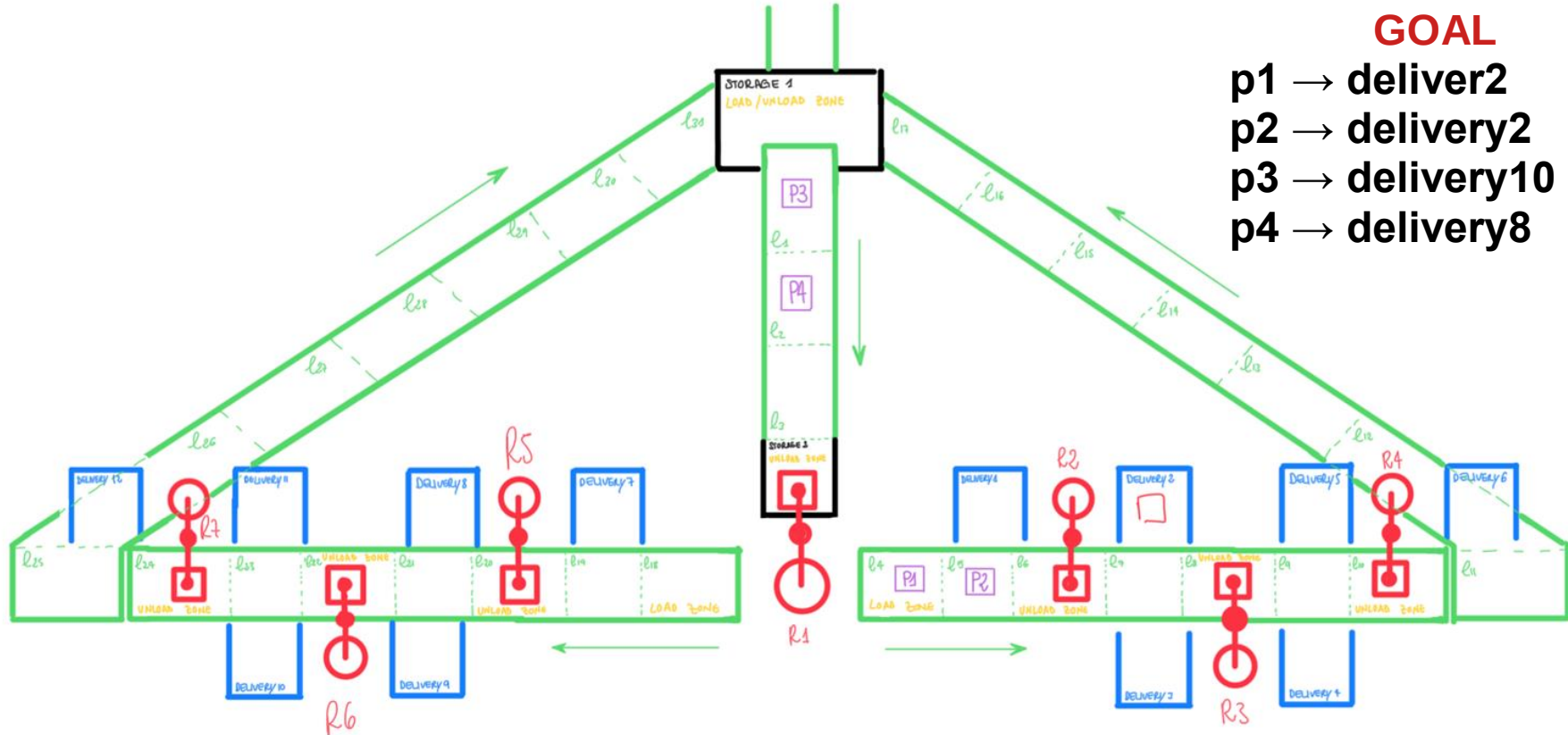
# PROBLEM 2



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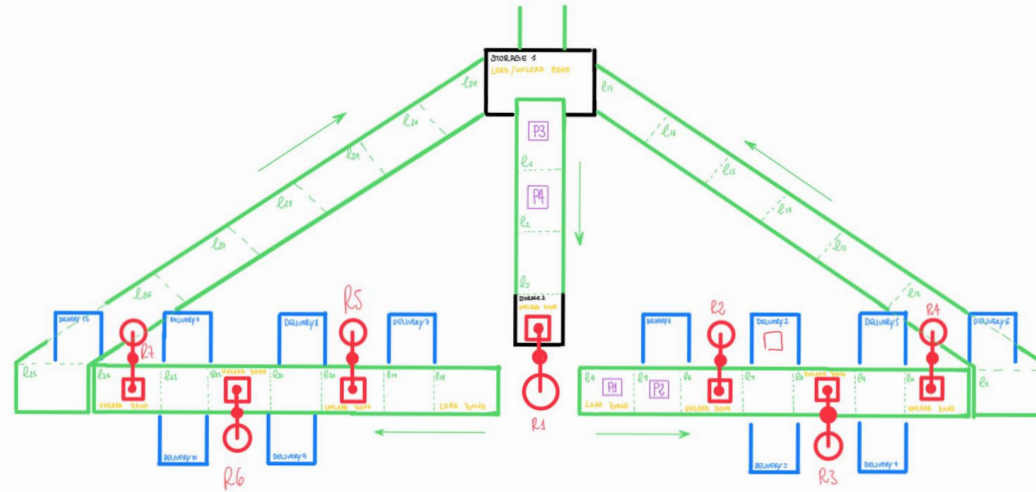
## GOAL

$p1 \rightarrow \text{deliver2}$   
 $p2 \rightarrow \text{delivery2}$   
 $p3 \rightarrow \text{delivery10}$   
 $p4 \rightarrow \text{delivery8}$





```
5.0: (intercept r2 p1 l6 cRight)
5.0: -----waiting----- [7.0]
7.0: (grasp r1 p4 storage2 cMain)
7.0: -----waiting----- [8.0]
8.0: (manipulate-with-package r1 p4 storage2 l18)
8.0: -----waiting----- [10.0]
10.0: (load-conveyor r1 p4 l18 cLeft)
10.0: (manipulate-no-package r1 l18 storage2)
10.0: -----waiting----- [13.0]
13.0: (grasp r1 p3 storage2 cMain)
13.0: -----waiting----- [14.0]
14.0: (manipulate-with-package r1 p3 storage2 l18)
14.0: (intercept r5 p4 l20 cLeft)
14.0: -----waiting----- [15.0]
15.0: (manipulate-with-package r5 p4 l20 delivery8)
15.0: -----waiting----- [17.0]
17.0: (deliver r5 p4 delivery8)
17.0: -----waiting----- [34.0]
34.0: (load-conveyor r1 p3 l18 cLeft)
34.0: (manipulate-no-package r1 l18 storage2)
34.0: -----waiting----- [36.0]
36.0: (grasp r1 p2 storage2 cMain)
36.0: -----waiting----- [37.0]
37.0: (manipulate-with-package r1 p2 storage2 l4)
37.0: (manipulate-with-package r2 p1 l6 delivery2)
37.0: -----waiting----- [39.0]
39.0: (load-conveyor r1 p2 l4 cRight)
39.0: (manipulate-no-package r5 delivery8 l20)
39.0: -----waiting----- [43.0]
43.0: (intercept r6 p3 l22 cLeft)
43.0: -----waiting----- [44.0]
44.0: (manipulate-with-package r6 p3 l22 delivery10)
44.0: (deliver r2 p1 delivery2)
44.0: (manipulate-no-package r2 delivery2 l6)
44.0: -----waiting----- [46.0]
46.0: (deliver r6 p3 delivery10)
46.0: -----waiting----- [75.0]
75.0: (manipulate-no-package r1 l4 storage2)
75.0: -----waiting----- [77.0]
77.0: (grasp r1 p2 storage2 cMain)
77.0: -----waiting----- [78.0]
78.0: (manipulate-with-package r1 p2 storage2 l4)
78.0: -----waiting----- [80.0]
80.0: (load-conveyor r1 p2 l4 cRight)
80.0: -----waiting----- [84.0]
84.0: (intercept r2 p2 l6 cRight)
84.0: -----waiting----- [85.0]
85.0: (manipulate-with-package r2 p2 l6 delivery2)
85.0: -----waiting----- [87.0]
87.0: (deliver r2 p2 delivery2)
```



```
Plan-Length:171
Elapsed Time: 87.0
Metric (Search):298.0
Planning Time (msec): 177759
Heuristic Time (msec): 154477
Search Time (msec): 177754
Expanded Nodes:1315680
States Evaluated:1724374
Number of Dead-Ends detected:561267
Number of Duplicates detected:1891206
```

The computation time is high (177s), but the exploration of about 1.3 million nodes allowed to discard 560 thousand dead-ends.





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## SEARCH LOG

```
H1 Setup Time (msec): 22
h(I):720.0
g(n)= 19.0 h(n)=702.0
g(n)= 25.0 h(n)=300.0
g(n)= 43.0 h(n)=209.0
g(n)= 58.0 h(n)=185.0
g(n)= 117.0 h(n)=139.0
g(n)= 142.0 h(n)=126.0
g(n)= 145.0 h(n)=125.0
g(n)= 162.0 h(n)=69.0
g(n)= 177.0 h(n)=29.0
g(n)= 180.0 h(n)=22.0
g(n)= 200.0 h(n)=18.0
-----Time: 10s ; Expanded Nodes: 85021 (Avg-Speed 8502.0 n/s); Evaluated States: 115529
-----Time: 20s ; Expanded Nodes: 166277 (Avg-Speed 8313.0 n/s); Evaluated States: 224779
-----Time: 30s ; Expanded Nodes: 248244 (Avg-Speed 8274.0 n/s); Evaluated States: 324378
-----Time: 40s ; Expanded Nodes: 317599 (Avg-Speed 7939.0 n/s); Evaluated States: 418226
-----Time: 50s ; Expanded Nodes: 392463 (Avg-Speed 7849.0 n/s); Evaluated States: 509856
-----Time: 60s ; Expanded Nodes: 474588 (Avg-Speed 7909.0 n/s); Evaluated States: 604724
-----Time: 70s ; Expanded Nodes: 543654 (Avg-Speed 7766.0 n/s); Evaluated States: 705117
-----Time: 80s ; Expanded Nodes: 607879 (Avg-Speed 7598.0 n/s); Evaluated States: 791116
-----Time: 90s ; Expanded Nodes: 687548 (Avg-Speed 7639.0 n/s); Evaluated States: 904985
-----Time: 100s ; Expanded Nodes: 767170 (Avg-Speed 7671.0 n/s); Evaluated States: 1003719
-----Time: 110s ; Expanded Nodes: 847625 (Avg-Speed 7705.0 n/s); Evaluated States: 1100855
-----Time: 120s ; Expanded Nodes: 926639 (Avg-Speed 7721.0 n/s); Evaluated States: 1205863
-----Time: 130s ; Expanded Nodes: 998456 (Avg-Speed 7680.0 n/s); Evaluated States: 1301889
-----Time: 140s ; Expanded Nodes: 1069660 (Avg-Speed 7640.0 n/s); Evaluated States: 1392927
-----Time: 150s ; Expanded Nodes: 1151823 (Avg-Speed 7678.0 n/s); Evaluated States: 1491467
-----Time: 160s ; Expanded Nodes: 1229945 (Avg-Speed 7687.0 n/s); Evaluated States: 1594857
-----Time: 170s ; Expanded Nodes: 1287036 (Avg-Speed 7570.0 n/s); Evaluated States: 1678888
g(n)= 279.0 h(n)=16.0
g(n)= 295.0 h(n)=1.0
Extracting plan with execution delta: 1.0
Problem Solved
```



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**THANK YOU FOR YOUR ATTENTION**